

Bone and Mineral Metabolism

ORF34-04 - Association between Semaglutide and Risk of Bone Fractures in Type 2 Diabetes



Sunday, June 14, 2026



2:00 PM - 2:15 PM CT



Location: Room W185ABC

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Oral Abstract Presenter(s)

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Introduction:

Weight loss interventions in patients with type 2 diabetes (T2D) can increase fracture risk through accelerated bone loss. Semaglutide, a glucagon-like peptide-1 receptor agonist (GLP-1RA), aids weight loss and may positively affect bone health, yet direct comparisons with other GLP-1RAs and non-surgical weight loss therapies are limited.

Aim:

This study aims to evaluate fracture incidence and body mass index (BMI) changes among T2D patients treated with semaglutide compared to those treated with dulaglutide or alternative weight loss drugs (phentermine/topiramate, bupropion/naltrexone).

Methods:

We conducted a retrospective cohort analysis using the Atropos Health Eos electronic health record dataset representing 161 million patients seen in community hospitals and academic medical centers across the US from January 2016 to December 2023. Using ICD9/10 codes, RxNorm codes, and LOINC codes, we selected adults aged ≥ 18 years diagnosed with T2D, with no history of prior fractures or use of osteoporosis medications. The intervention group received semaglutide ($n = 26,324$), and control group received dulaglutide, phentermine/topiramate, or bupropion/naltrexone, without prior history of semaglutide use ($n = 33,555$). Key outcomes included baseline BMI (3 months pre-index), 1-year BMI (9-15 months post-index), BMI change (delta), and post-index fracture incidence. Analyses utilized high-dimensional propensity score (hdPS) matching.

Results:

After hdPS matching, baseline characteristics were similar between the semaglutide group (n=17,506) and the control group (n=17,506): mean [SD] age, 54.1 [12.4] vs 54.5 [12.7] years; female, 55.4 vs 53.9%; and BMI, 38.6 [7.8] vs 37.8 [8.0] kg/m². In the subset with available BMI data (n=4,191 pairs), semaglutide treatment was associated with greater reduction in BMI compared with the comparator group (delta BMI −1.9 [3.5] vs −1.2 [3.6] kg/m²), with a sustained 1-year mean difference (−0.72, 95% CI −0.87 to −0.57; p< 0.001). After mean follow-up of 1327 [1903] days, semaglutide group had 794 fractures compared to 1045 fractures in the control group (HR 0.85, 95% CI 0.77–0.93; p< 0.001).

Conclusions:

In T2D patients, semaglutide was associated with a 15% reduction in fracture incidence and greater weight loss compared to another GLP-1RA and alternative weight-loss therapies. These findings highlight potential bone-protective effects of semaglutide, though prospective studies are needed to confirm findings.